This transcript was produced from a phone recording of a subject with an accent. As a result there are more "_____"s and "<inaudible>"s than in a normal transcript. -- Production Transcripts

KT: So [redacted], you agree to be recorded?

Interviewee: Yes, I do agree to be recorded.

KT: And this is [redacted], I agree to be recorded.

KP: And this is [redacted], I agree to be recorded.

KT: Right. And then [redacted], you've had a chance to review the consent and you consent to being interviewed.

Interviewee: Yes, I consent to be interviewed.

KT: Obviously because you're on here. And as we discussed, we will take a shot at preparing a redacted transcript, present that to you, you will review it, decide whether it has been anonymized sufficiently and whether there's any material that needs to take out of it to post it or whether it be best that we don't post it at all, entirely at your discretion and that will be done probably sometime in the late spring, early summer. So diving right in, what we'd like to get to today is get some background material just on the record for you and your experience in research that's related to genome editing and on the bioinformatics side, what you're currently doing and then really dive into what your interactions were with the FDA. I think the most important things for us are how the develops in the bioinformatics area in a sense came to the FDA, how you were recruited to the FDA, what you developed while you were there, what would be useful for and other observations about your experience. So starting, we have some questions we'll just walk through and move through that and you can just freely respond. Can you just give us a little bit about your educational background and how you got involved in the research that you do that's relevant to gene editing, the bioinformatics?

Interviewee: Okay. So by training I am a [redacted] , that's how I was involved with large scale computing systems [redacted]…

KT: Let me just interrupt for one quick second. So what was roughly the timeframe?

Interviewee: I went to [redacted] in 2000, [redacted] and then I went to [redacted] 2010, 2011 timeframe, somewhere…

KT: About there. And at this point they were using the next gen sequencing to analyze the genetic impacts of some of the therapeutics that they were reviewing for IND clinical trials and marketing and so forth.

Interviewee: Two types of scenarios and one of them was using next generation sequencing as an evidence for in that particular case resistant development in an antiviral case, that was the very first NGS based submission, it's a published information, you can read about [redacted] FDA looked at the clinical data, requested raw data and raw data were big enough for FDA to [redacted] not have enough technology, [redacted].

KT: Oh let me just ask a little-- so that just because we're not that well-- so these are very complex software programs that then can take large amounts of data and then analyze them [redacted].

Interviewee: Let me kind of give you <inaudible 00:06:33>.

KT: Yeah, that would be helpful.

Interviewee: So when you have a computation to perform, usually you run the computer program in your single computer and the entire operation happens in the memory of a single computer and it does the analysis and produces the outcome. The data is so big, it doesn't fit into one computer and then that means you have to split the computation and do it in thousands of computers and then the second thing, the complexity is so big, in one computer even if you fit the memory, it will take you 50 years to finish one computation, so you need thousands of computers to do parts of the job so you can finish in a realistic amount of time. So it's easier said than done, we have to parallelize [ph?], yes. But then other than the very logic of your algorithm which is trying to compute the result, now you have a logical interacting pieces of the computation, yes, and that information has to flow, instead of coming to one place, information is distributed along the computing nodes and in each piece computes its part but then there are interactions within different computing units that they have to do. Imagine if you are working for an office and then there's a big enough job that you need 40 employees to do, yes, if the job is easily splittable so each one can do independently their part, that's called embarrassingly parallelizable algorithm. If the job is complex, then when parts, 40 employees work, they have to constantly interact to each other and messages and information has to flow between them, that's called complexity parallelizable algorithm. So 80 to 90 percent of algorithms are complexity parallelizable algorithms, they're not embarrassingly parallelizable and you need an extra layers of logic and a huge infrastructure to organize that flow of the data, logic and processes. That's what [redacted] is.

KT: Let me just step outside for a second and say, thank you very much, that's a brilliantly clear analysis and explanation, I was very worried about are we going to be able to get a handle on this and this is quite, quite helpful. And so just to close the loop then when you're dealing with analyzing genetic data, the genetic data is this nature of it's very, very complex and its interactions and how it is performing in the body becomes more complex and these are the need for these sort of multiple nodes and the complex coordination among them.

Interviewee: It's large and it's complex and it's multidimensional. So large because information about each patient is huge, I can describe the size of the data in a little bit, it's multidimensional because it reveals different natures of information about you, it's not just [redacted] doing the same thing over and over again, [redacted] doing different things on different aspects interacting. A genome is not just big line of string, genome has genes which concurrently produce different phenotypes of behavior or cell, so

interaction is multidimensional, it's not a one long line, it's the volume of the information, information it's not linear despite the fact that it's actually allocated in a chromosomal like when it starts executing different parts of it execute at the same time, it becomes the informational volume of different dimensions doing something, functional dimensions of so many, yes, so that's why you need-- it's a sophisticated and it's very large and it's multidimensional algorithms.

KT: And the challenge for the FDA [redacted] was the [redacted] that you said there was first clinical data, the FDA wanted some raw underlying data that required the use of sophisticated genetic sequencing and in order to analyze and understand that sequencing the tools that they currently had didn't seem adequate [redacted] in order to do that.

Interviewee: Tools and infrastructures and the size is very important. Let me give you an example, it sounds like a joke but it is not, so at some point, this is a [redacted] submission, another submission, the company won't have to submit genetic data and one of the old style regulatory officers was annoyed with the fact they have to allow newer types of data to flow into existing system which is very inertial sometimes, yes, and then [redacted]. He said, "Can they print that and send us the paper?" [redacted] for one file, a single file and we have hundreds, we needed ten 18 wheeler trucks full of paper just one file. [redacted]."

<laughter>

Interviewee: [redacted] The next one [ph?] is very large, it doesn't fit in computers. Then let's step back, now you have many computers, that means you have to buy many computers, you have to order ahead of time, there is contract vehicles to deal with, now you have to connect the computers into a network, that means you have to think about networking, firewalls, security, all of those aspects and then you are putting proprietary data, that means security levels and implementational controls, who has access to this room, who has system access, who has a root access, who can do what because there are different types of people. Immediately a single computer computation becomes an infrastructure problem. Now you have the layers of people who have to work with the system to wrap a one computation with networking, physical location, air conditioning, all of that stuff, it's amazing, [redacted].

KT: And this was at the FDA.

Interviewee: Yes, at the FDA, yeah.

KP: I was going to ask, so just to be clear, [redacted] FDA [redacted]did not have a datacenter to do any of this, [redacted]

Interviewee: There was a datacenter which was only fraction of the space, a fraction of the power [redacted].

KP: Okay, but you built…

Interviewee: So that datacenter didn't have the scale magnitude and then it didn't have the new types of data capacity, it's a newer type of genomic data, I mean the entire NGS story started in 2005, 2009, yes, FDA at that time was not ready for the scale and the types of algorithmics of genomics, yes, there were a few expertises of bioinformatics, people could do small things on their computers, yes, or in some clusters outside but, hey, hey, hey, this is FDA, you cannot take the proprietary information outside, it has to be inside, yes, NIH had a big resource, they could do NGS but that was not, FDA cannot move [redacted] data to NIH in a public computer to compute something because if that happens, the resource, public resource of NIH is a shared space. Let's say I am FDA, I do proprietary type information computations and you are a researcher who is doing his public research, yes, if my computations run on that compute unit and pieces of information are in memory, your computation can allocate the memory and before writing into it, start reading it and ahead [ph?] I just lifted information proprietary to you, that is not allowed, that's why FDA had to build their own ecosystem which is running on a segregated hardware.

KP: Okay. So if I understand, the issue was that they didn't have the hardware in-house and for compliance and confidentiality reasons, it cannot go out of house, [redacted].

Interviewee: [redacted].

KT: So now this is occurring prior to the invention of CRISPR Cas genome editing…

Interviewee: Of course.

KT: …and the need for the next NGS, the next generation sequencing is because of being able to analyze the data that was done in regulatory submissions that weren't genome editing regulatory submissions but that they were in order to understand the impact at a molecular level of a therapeutic such as the [redacted] drug.

Interviewee: And/or provides manufacturing consistency control sometimes.

KT: Oh, so CMC related.

Interviewee: Yes, let me bring you an example, so you are a company who produces attenuated virus as a vaccine, yes, what it is, you take a virus, you put it into environment that it is forced to change and while it changes it loses its pathogenicity, yes, so that is a vaccine now because given to you, your body has the time to kill it, recognize and kill it before it kills you. But then we are cooking these things continuously, sometimes that this is happening in big containers, somebody changes the material of the valve, slightly increases the number of <inaudible 00:17:26> or some temperature variation, virus adapts to it and gains _____ 00:17:31 and now it can kill you because it's pathogen, you think you have a vaccine but you have a vaccine, yes, and the easiest way to monitor is not to use it on monkeys which was done before but to check genomically if that mutation is back or not.

KT: So the next gen sequencing was used for basically to evaluate all the biologics that are being produced in bioreactors and to make sure you have manufacturing consistency which is one of the major challenges in the develop and the rollout of biologics.

Interviewee: You got it.

<overlapping conversation>

Interviewee: Diagnostics is the other type of a use case. Let's say, I'm going to give you an example, lettuce, people start getting sick from lettuce, yes, so you want to find out where is the lettuce coming from, yes, so are you going to stop the entire American import of lettuce, no way, you can't stop the borders, cannot close the country's economy, it can-- what happen now, so the way FDA does, they build the global map of food borne pathogens, yes, so now if I get the lettuce and I know which pathogen is there I can know that, "Hey, I have to close the shipments from particular port in India and not stop the entire lettuce manufacturing in California or Brazil."

KT: Right, ah...

Interviewee: Again, if you think of the genome of a lettuce, genome has a fingerprint in it, I can recognize exactly and pinpoint where it's coming from, again you need to know the genome.

KT: So what happened was the emergence of NGS created all these capacities for the FDA and in order to have the infrastructure and the tools to exploit that capacity was the source of developing this project and they couldn't go outside because [redacted] of the proprietary elements that they were doing for regulatory review purposes.

Interviewee: Yes.

KT: So how were you identified, were you known, had you worked with the FDA while you were at NIH or…

Interviewee: [redacted].

KT: So the seed for this was a medical grant or medical…

Interviewee: [redacted].

KT: I'm sorry?

Interviewee: [redacted].

KT: So this was something that the FDA had some funding that they could start doing an analysis of. So at any point did you get involved with any of the genome editing or the gene therapy analytics?

Interviewee: The first [redacted] was done at FDA. [redacted] team actually have built the ecosystem for [redacted] data, we had to build-- well before we go into it, let me say something which is to understand the nature of the work and then when I make comments about the project you'll understand underlying things. So the team is a research and development team, [redacted] team, so we get a call, [redacted] to

participate in a pre meeting where genomic evidence might come so we go there and then obviously companies are developing a project for that and they have hired bioinformaticians, data analysts, collected the data, they have algorisystems [ph?], they are doing something, [redacted]. And in the three weeks to one month time come up with a production quality software which is not worse than what the companies had as a result of years of work. So you understand the stress level.

KT: Right. So the companies have their own proprietary software programs that do the data analytics internally, they present the results to the FDA but the FDA needs a mechanism to check that accuracy and so forth and that's why they need their own system and these are not things you can buy off the shelf, they're things [redacted] have to customize.

Interviewee: Yep, the technology changes so fast and the algorithms come up every day so FDA needs to stay relevant in this age of change, FDA needs a constant R&D support, constant.

KT: And where do they get the resources for doing that?

Interviewee: Well, I mean they find the projects like [redacted], I mean today, every year, [redacted], FDA was funding into [redacted] platform, yes, I mean in some years it would be more than that but in average I think it's three million dollars a year resource and then we pretty much had the green light of hiring the best people in the world, that's it, and then we could have any hardware we want, but at the same time, we helped so many products come to market that it was all paid for many, many times.

KT: Paid for in the sense of the [redacted]

Interviewee: No, no, no, paid for-- I mean obviously…

<overlapping conversation>

KT: Social benefit.

Interviewee: I mean the spending was completely justified, that's what I meant.

KT: Yeah, in terms of the resources within the FDA. So I just started to ask, were you involved in any projects that had to do with the genome editing or gene therapies?

Interviewee: Yes, the first [redacted] products when they came, similarly we were called to design an ecosystem for accepting [redacted] data, yes, and analyzing the [redacted] data and then we actually-- that's why I remembered to tell him this thing, yes. So yes, I mean we did the first [redacted] and then all other-- I mean there's not many gene editing products which have gone through FDA.

KT: Right, exactly.

Interviewee: But there are so many preliminary research projects with the FDA already where the involvement of the gene editing and development of the strategies, how do you evaluate gene editing

products that have been done in [redacted] team already, we developed certain-- obviously FDA mandated us to develop a gene editing evaluation software and then we have built that already and then we have done research projects with that and then already review projects as well. Where, you know, in gene editing, I mean-- you know the technology or you want me to kind of…

KT: I work with people who do it so I'm fairly familiar with the genome editing technology.

Interviewee: Okay. So submissions [ph?]of on target and off target are very important, what you intended to change, it has to be <inaudible 00:25:41> you succeeded and what you did not intend to change, you have to prove it hasn't changed or the changes that may have happened they do not have an impact, functional impact on the life of a cell and the patient as well so that on target, off target is a significant issue and then human genome despite us studying it for so long, we still have no 100 percent understanding of what is the stature [ph?] of genome per patient. About 30 percent of the genome can be arguably assembled into different scaffolds, so understand we are trying to figure out if the gene editing has altered part of the genome which are not in the conservative regions of the genome, that's a big task, yes, that off target is very big task. Also off targets can be functional, off targets can be chromosomal, yes, I mean you might-- let's say if you change a place in genome which is not a coding region and that intergenic region doesn't play any functional role then maybe and also if that gene you accidentally change is not expressed in the tissue that you targeting to modify, that might not be a big issue. [redacted].

KT: So the FDA is receiving the NGS data from the drug sponsor and the drug sponsor says, "Oh, here's the off target effects, they're fine, it's not a big deal, here's what we found," and then what the regulators have to do is to sort of take a second look and see…

Interviewee: Yeah.

KT: Yeah. So they're not doing the actual analysis of the off target, I mean the location of the off target effects…

Interviewee: We do.

KT: You do. So do they also receive samples of the...

Interviewee: No, no, no. Usually if it's in GS [ph?] evidence we will receive the pass-through files [ph?] of whatever the regions have done [ph?] if it's a new [ph?] genome, or it depends. I mean, it depends on what type of data we receive, yes? You have to build your pipeline off of data. I mean, if it's a type of an optical sequencing or some sequencing, whatever they have they send us, we build the pipeline on top of it.

KT: So they send you the data about the sequencing, and then the FDA has to see if the data discloses off-target changes that actually are problematic that the sponsor overlooked or was unaware of or whatever.

Interviewee: Yes.

KT: So that's...

Interviewee: Yes, yes. You are absolutely right, and actually sometimes if FDA considers that's not enough information the FDA might request back some more data, so "Okay, I mean, the guide seq is out [ph?]. Why don't you do guide seq, send us the data to prove your point" and things like this, yes. I mean, we were trying to at the FDA minimize potentially not noticing the adverse events on the genome, but the FDA at the same time has to try to maintain minimum hindrance. You can be comprehensive to a certain degree. Otherwise no products will ever come to market. You have to prove that you are not creating a product which is dangerous for a human being.

KT: Right. The layperson might say "I get the file. The FDA just looks at the file and says `Okay, it looks like everything's okay,'" but you're saying that file in a sense would be five 18-wheelers double-sided, and it's not just looking at the file. You actually need the program and the infrastructure to take that electronic file and process it and say "What's the data saying about potential off-target effects having adverse impacts?"

Interviewee: Yes. And, I mean, technology is new. It's agreed [ph?]. I mean, you don't have the long safety profile for it. We have a long historical safety profile for certain protein products, yes, or the blood products or some cell therapies, but this is a completely new technology. A level of scrutiny is still needed at least initially before we can prove and understand the safety profile of the product. Then maybe once we start trusting it then in 10 years from now maybe we don't need to validate every single step of the way <audio dropout> technology. Like blood plasma. We are giving an allogeneic product to another human, yes? I mean, that's dangerous. That actually is dangerous, but now we kind of trust it, and in T-cell therapies, immunoncology or immune edited genome injections you are giving a allogeneic product of very aggressive cells to another human being, yes? I'm assuming <audio dropout> allogeneic, because I see the future in allogeneics, not so much in...

KT: No, exactly, right.

Interviewee: So we will be giving an allogeneic immune-risky product to a human being. We have to prove the safety profile first. We have to create that framework, and then in five, 10 years after we do this for cancers or the rare diseases I wouldn't be surprised to have a topically applicable cream <audio dropout> so I can be a long-haired, curly, blondish man.

KT: <laughs> Right. And your point is that because of the novelty there is a need to do extra scrutiny and to balance the scrutiny with facilitating development and access.

Interviewee: And a long impact.

KT: What's that?

Interviewee: Have a long impact. They have a long life inside a human being. The way we monitor the product-- and it's very important-- in the human being-- and these perspectives I'm expressing now, those were all considerations when I was at the FDA reading scientific reviews of these products, yes, so let's say you inject a human being with modified T cells, yes? I mean, you are gonna have memory T cells which have a shelf life in a human being, and we are vaccinating a human from a cancer pretty much by having memory T cells of the gene-edited, let's say, cancer-aggressive markers, yes? So how do you monitor in a human being what's happening? When do they deplete completely the cells? Because the human himself is not gonna produce them anymore. It's only the ones you give, yes, so how do you monitor the population? And those are the inside fight of the NK cells [ph?] of the human being with injected allogeneic cells. You have to monitor that dynamic. We don't even know-- and this is my scientific opinion. Today we are still not there to completely understand that interaction.

KT: Right. I'd like to turn to [redacted] staffing [redacted] Let me step back for a second. You were an FDA employee, and the people you were recruiting were FDA employees, not outside contractors.

Interviewee: No, we hired people from outside labs, so there were two models. Now we are hitting one of the pain points, which-- I think it's important to have those pain points on staffing, yes? So a classical organization like FDA at that point did not recognize how valuable bioinformatics is. I mean, obviously certain people did. That's why we were funded [ph?] [redacted], but a lot of time it was ignored, like "Bioinformatics is just this bunch of software guys who are running a bunch of tools," yes, and then this expertise hasn't been valued that much at that point until we brought to the attention that unless this bunch of software guys are doing this bunch of things we are not validating products anymore. We are irrelevant. So it took us a few years to prove the point, but then FDA is a organization which classically hires statisticians, yeah, doctors, and the statisticians and clinicians most of the time, so having a bioinformatics core at the FDA was not something which was natural, yes, for them. <audio dropout> exactly what we needed here, so most of the FDA full-time employees were not bioinformaticians, [redacted]-- there was not even a job description in the federal government which would fit for a bioinformatician. [redacted] had to hire through what is called ORISE fellowship program. It's a Oak Ridge Institute fellowship program that's a-- you hire young people which are less than five years from their education who've finished some education. It's a program which is supposed to benefit them and kind of teach them in regulatory sciences, yes, all of these things while you are allowed to use them for research and development influence [ph?], so that's how we started, but how do you get world-class experts into this? Yes? <laughs> <audio dropout> permanent job, [redacted] so many bureaucratic fights of allocating full-time employee locations, employee spots and things and convince these bright and talented people to stay with us until we put them into full-time employment, [redacted] and it was really bureaucratically not easy, but we did it. FDA kind of at the end...

KT: [redacted] [ph?], were you starting to say something?

KP: I wasn't. You can go ahead.

KT: So through the ORISE program [redacted] able to recruit junior people and then [redacted] still needed senior experts.

Interviewee: [redacted].

KT: [redacted] that took convincing the FDA to create positions and then recruit people into the positions who were senior. I take it you were saying that was a process of years more than months or many months?

Interviewee: [redacted]. ORISE was limiting five years after education. [redacted] through exceptions with the leadership of ORISE and the FDA to create exceptions to get few experts into the program and then...

KT: Into ORISE.

Interviewee: ...that gave us few years [redacted]. I mean, it was learning [redacted] in an FDA environment to work [ph?].

KT: [redacted]. When they were in the ORISE program were they temporary FDA employees?

Interviewee: They are not FDA employees, and they couldn't touch regulatory data.

KT: So they were consultants or postdocs but not...

Interviewee: Yes, postdocs and things, and then only the employees could touch regulatory data.

KT: [redacted] they would have to create a position, recruit for the position, hire someone into the position, and at that point they could go...

Interviewee: [redacted].

KT: You worked also with FDA researchers who were not the regulatory people but people who were doing bench research?

Interviewee: About a third of FDA is actually scientific research at CBER at least, yes, and about quarter I think at CDER or something, so, yes, I mean, there were a large number of projects. At some point we had more than 80 project going in height [ph?]. That's very good number. I mean, we published 40-something publications ourselves, and another 80 publications came out [redacted]. I mean, that's a good record. In 10 years more than 100 publications came out, 10 publications a year of that complexity. [redacted]

KT: Do you know how people developed their research programs? Were they doing research activity that was requested by the regulators?

Interviewee: Yes. So, first of all, I believe,[redacted], that you cannot be a good scientific reviewer if you do not have your own research, if you are not on top. You have to be on top, yes? You have to be one of the pillars of that industry in order to be able to evaluate, because that's the most responsible job anybody

can have, evaluate drugs to the market [ph?]. So, yes, FDA has significant funding for internal research programs. They are usually oriented towards massive development that will facilitate either drug development or development <audio dropout> evaluation of drugs or devices or biologics, yes? So there is a large bunch of money in CBER and CDER and CDRH for this type of research, so there are intramural grants, and everybody could apply. I myself got [redacted] grants funded by FDA, yes, and then you could apply for a grant, you could receive a grant. You could be also part of the NIH grant, although you could not be PI of an NIH grant, yes, but then you could have an interagency-type agreement with NIH researchers to transfer the money between them, so yes.

KT: So these researchers would also be people who were in the regulatory, so they were research-slash-regulators. Some of their time they spent doing the research, some of the time doing the regulation. [redacted] who was not FDA employees, could work with these people when they were doing research but not if they were using data from a regulatory submission.

Interviewee: [redacted]

KT: [redacted].

Interviewee: Yep.

KP: [redacted]?

Interviewee: Yeah, yeah, I can. So, I mean, if you want I can even demonstrate if you are interested, I mean, but not the FDA one. I cannot show you the FDA's, <laughs> but I can show you some of the other things if you want. I mean, I can make a demo for you if you are interested about that. Are you interested, or...

KT: I find it quite fascinating. I don't want to overburden you.

Interviewee: No, so you're not overburdening me. Hold on. Give me one second. I'm gonna open one of the computations, and then I'll share my screen with you. So how does it look like? Yes, and to make sure this is not-- we are not gonna show FDA. [redacted]?

KT: Right.

Interviewee: I am disabled to share screen. Can you enable me?

KP: Sorry. I think that should give you the ability to share a screen.

Interviewee: Okay, so this is one-- hold on. Let me...

KT: There we go.

Interviewee: ...get the things on [ph?]. Hold on. [redacted]?

KT: Right. It's fascinating.

Interviewee: <overlapping conversation> [redacted]...

KT: Right. This is fascinating. So it provides a tool to give them the ability to do this complex sequencing work without having the underlying skills.

Interviewee: Yep, [redacted] So this is how it looks like, yes, so that's why it's easy for the scientists to work with a system like this.

KP: So at FDA someone who's been trained how to use the interface but not trained on how the back end works can still run functional and useful analyses of data to make decisions about verifying data.

Interviewee: Absolutely. Absolutely, because alternative is to run everything manually. That's a much more difficult process, and it involves shell screen, computer programming, etcetera, large files moving and things, and some things you can't even do it manually because there are thousands of computers, and you have to kind of parallel this. You don't do it manually. <laughs>

KP: From what you showed us it's relatively real-time, maybe a 30-second or one-minute wait.

Interviewee: <overlapping conversation> Absolutely. It is real-time, yes. <overlapping conversation>

KP: I'm sorry?

Interviewee: It's a small genome, influenza. That's why it finished real-time. Human would take about 15 minutes.

KP: But not like 10 days but like 15 minutes. Okay. Did that time scale change over your time at FDA [redacted]?

Interviewee: Well, I mean, obviously, [redacted].

KP: So basically the majority of the work now has shifted from waiting on compute time to coming up with what to ask the computer.

Interviewee: Yes, exactly. Perfect.

KP: That was not true when you started off.

Interviewee: Yes, but for human-- let's say we have to make a decision if a human should take panitumumab or not. That's a $12,000-a-month drug, so you don't want to get the drug if you are resistant to it, and we know about 60 to 70 percent people are resistant to it, so that computation if I do his genome and, say, deep enough and then I get it I can answer that question for a single human in about 25 minutes.

KP: So clinical speeds essentially.

Interviewee: Yeah, yeah. <overlapping conversation>

KP: I'm sorry?

Interviewee: The computation is much faster than sequencing itself.

KP: You said way back at the beginning that FDA suddenly realized with NGS data that they needed some bioinformatic capacity [redacted]. Can you talk about how that became a priority and how that developed?

Interviewee: [redacted].

KP: You mentioned there were challenges to get FDA onboard with bioinformatics. [redacted]...

Interviewee: [redacted] when I say FDA didn't understand I mean FDA is very heterogeneous organization.

KP: Of course.

Interviewee: <overlapping conversation> the FDA are top-notch scientists. They are top-notch scientists publishing, doing excellent work, etcetera. These guys had never issues actually to understand that bioinformatics is important because so many times they had a data challenge they didn't know how to solve it. So one-third is just an ecosystem support. That's not relevant to discuss here, and the other third is the administrative people of different layers, yes? When I say they did not understand it doesn't mean the entire FDA. It's a few organization, because think about this. People thought data is IT. IT belongs to IT department. IT people, they are not scientists, yes, so in a way data would go under the handling of the IT, and those are programmers, developers. They understand how to secure things, not how to compute things, yes, and they understand how which ones should be connected to which person to which active directory element [ph?], but they don't know what the next genome sequencing is, and [redacted]."

KT: <laughs>

Interviewee: [redacted]."

KT: <laughs>

Interviewee: [redacted] Yeah, so, but, I mean, you have to understand FDA is a very serious organization that has to protect itself. There are millions of automated attacks a day to FDA, and then there are few tens [ph?] of coordinated attacks on FDA today because strongest keeps attacking [ph?], so you should accept that, God forbid this information gets into open public domain, quarter of American economy will be under danger of having no protection of information, yes, so you don't want to do that. But, I mean, eventually things got work [ph?], so IT people originally didn't understand the need for

bioinformatics. They thought that we just copy the software and it runs, and we don't need bioinformatics, but every pipeline design is bioinformatics. Every new custom software design is new bioinformatics. I mean, once I explain what is the alignment-- you know what the alignment is, yes? We don't have a technology which reads the human genome from left to right, so we shred it and then read the pieces, and then we have to find out where the pieces go, yes? That's called alignment. [redacted]

KT: It sounds like the core skills came in through the ORISE program. If you were head of CBER or deputy or commissioner, do you think there could be better ways to pull bioinformatic skills and expertise that would be relevant to the mission of the FDA?

Interviewee: Yes. I mean, first of all, I would not like to be in [redacted] shoes, because [redacted] has so many other things to worry about, [redacted] , CBER director. He's a very good person, but his shoes are too much. So I think what I would do is make sure that science is relevant to the stench where you're living [ph?], yes? So that's important. We live in a data era. We live in a era of data. Remember-- now, this my son told me actually. He's a very intelligent young boy. He said "The first tools were made of gold, yes, and then because they were shiny pieces of metal and things, so we were in a"-- no, stone. Made of stone first, yes? "Everything was made of stone. That's why it was called Stone Age. Then we discovered gold and the things, and we started making tools. It was the Golden Age. And then Iron Age because we started making tools of iron, and then Industrial Age, etcetera. Today we are Age of Data, because our tools are made of data, and what allows us to build things today is the data." And in an age of data you have to have scientists, you have to have enough resources allocated to do data science, yes? It's no different from having metallurgist in 1900s. It's no different from having stone workers 2,000 years ago. It's an age of data. I think it needs to be funded better. I truly believe that FDA and all other organizations allow having bioinformatics as true part of the workforce, and that's not what it is even now.

KT: What do you see as the impediments that restrict the FDA from making these changes?

Interviewee: Well, I mean, old legacy systems are difficult to change, yes? I mean, there's inertia in these big systems. Inertia is needed for an organization which has to be not too dynamic, because every wave it comes to it shouldn't change. At the same time, I mean, you have to constantly change with informatics. I mean, impediment is that-- ahh. It will be constantly more and more difficult for FDA to stay relevant if they don't fund bioinformatics and data science. It will be more and more difficult. FDA has to start trusting more than verifying, and that's not FDA mission, yes? I mean, you have to verify first. Trust must be earned, but if you have no other tools and you have a timeline you have to trust.

KT: You're saying that the data scientists are necessary for the verification, and in the absence of a robust enough verification capacity you're reduced to trust.

Interviewee: Yes, you are reduced to trust, and that works when it's a platform which has a long history. That doesn't work for situations like this. I mean...

KT: Like genome editing and new novel genetics.

Interviewee: Yeah. I can give you an example. I mean, when we do the analysis never in any single analysis we came to identical conclusion with the sponsor. I mean, there were always differences, and in certain cases the differences were admittable. It's okay. The differences in our computational outcomes are not significant enough to change the meaning of this evidence, yes, but in third of the cases I would say we came to a conclusion that either resulted in label change or the refusal of the product or something else.

KT: So you're saying that the bioinformatics analysis that was done frequently showed variance from the informatics conclusions that were presented, and in about a third of the total product review you found things that caused significant changes like label or a denial.

Interviewee: <overlapping conversation> You'll have [ph?] a label change, yeah, and that is significant, yes.

KT: Very.

Interviewee: Yes, and then, I mean, I'll give you an example with a big clinical trial, [redacted].

KT: Right.

KP: You were saying one of the things the FDA needs is to devote more resources in-house to bioinformatics. What does FDA ask for when running a bioinformatics verification? It sounds like they submit their raw data, and then you design and run your own analyses. Does FDA also ask for the analytic design? In political science you must specify your model, and you submit your raw data and your model, and people can verify. Is that standardized? Should it be? What are the requirements?

Interviewee: Yes. You are hitting something which [redacted] So...

KP: There's a black box.

Interviewee: Yes. [redacted]

KP: To be clear, this is not just a [redacted] standard. This is a [redacted] standard, [redacted].

Interviewee: [redacted]. So I'm gonna ask you a question. Have you ever drunk a ibuprofen?

KP: Yes.

Interviewee: Of course, yes. Why do you trust it?

KP: It's been around forever.

Interviewee: Because <audio dropout> protocol in its manufacturing.

KP: Oh yes. Sorry.

Interviewee: There is a fixed protocol, and no violations are accepted from manufacturing. It's not like some pharmacists are not creative enough. "Hey, I'm gonna add a little spice today," yes? So they are, but you cannot. It's prohibited. So a computational outcome is a product. It's a device. You have to have a protocol of manufacturing it fixed. In a science domain, no, you do what you want to do. The moment it becomes a product it has to be standardized, formalized, so its outcome cannot be changed. [redacted].

KT: [redacted].

Interviewee: Yes.

KT: We really appreciate your time, and this has been extremely informative, and we've run over. I don't want to take more of your time. Did you see how the FDA personnel you were working with would increase their skills and ability in this or other areas? You said they did research. Were there other skills development opportunities, particularly in the bioinformatics area, that you saw with your colleagues at the FDA?

Interviewee: Well, I mean, there was a lot of support for educational programs with relation to bioinformatics. [redacted] lectures around the FDA. Every month we would have a lecture. I mean, first of all, we had people whose only job was to teach reviewers how to use this, but also every month we would invite a technology or bioinformatics platform to come and present at the FDA, so we had pretty much [redacted], I mean, anybody who was willing to. We had a series of seminars on bioinformatics, and [redacted] top scientists to come and present their work and things like this in many different areas, yes? It's like in microbiology, oncology, virology, you name it, and it was very well-supported. Attendance was like every time I would make the meeting from 70 to 100 people would be in a room, yes, so they knew if it's a [redacted] serious [ph?] they would rather give them big room because there's not gonna be enough space [ph?]. <laughs> So, yeah, and then when they come [redacted] opportunity to train scientists, not just come present and then go. So actually [redacted] the companies to train scientists, like make lectures, I mean, with the room set-up full of computers, and everybody had created account for them. Everybody went up [ph?], clicked buttons and did stuff, not just [redacted] platform but on other computing platforms as well.

KT: So sort of an interactive seminar.

Interviewee: Of course, of course. I would say every third and fourth was interactive seminar.

KT: Did people go out and attend conferences?

Interviewee: Yes. Yes, of course. Yeah, you're right. I mean, you're asking all of the right questions. I think you know what you are doing. <laughs>

KT: Did you find that the junior people could go or just the senior people?

Interviewee: All of them. All of them. I mean, there were many conferences. [redacted] a budget to send [redacted] junior people to conferences, but also I was never myself or any of senior people refused conference-going. In fact, it was very well-supported, and FDA pays for all of the conferences FDA people were going, and you just need to make sure that it's relevant to what you do and a green light. It's important. Science is very well-supported at FDA. I mean, it is given, okay, so I have no complaint.

KT: Great. [redacted] [ph?], any other questions?

KP: No. I think this has been just beautifully informative. I said no, and then I have one more. I'm sorry. [redacted] Do you think there should also be standard assays, hurdles or tests that the FDA requires all submitters to submit with their new drug? Does it need to be standardized, or is it more important to focus on the protocol?

Interviewee: Yeah. So, I mean, obviously, I mean, my expertise is in bioinformatics and data science. Yes, I mean, there are standards for overall [ph?] submission, and there is a master file, and there is all type of standard for underlying file formats, and there are all types of standards, but what is not yet well there is that the provenance framework is not strong, and it's not just about the FDA submission. It's overall. I mean, I deal with situation when I work with biotech companies in pharma that samples are mislabeled. I mean, they claim these are two brothers. One is sick, one is not. We are trying to understand what it is which is different between them, why one gets sick and the other doesn't, and then I recognize they are not brothers. Samples are not from brothers. The labeling is not there. The labeling of the biospecimen is done separately from labeling of the files, and then the labeling is broken, and labeling is broken almost every second submission, samples, labeling, and the trajectory of the data and trajectory of the samples is broken. They are not well-connected. There is a human part there which should be removed. Humans are error-prones. I mean, I get submissions with 2,000 files in them. This file is from human number one, baseline, day zero, the drug not yet administered. This sample is from the same human day four after drug administration. This sample is from a different human, baseline, so on, so forth, and 2,400 files, for example, like one of the submissions I'm working on right now, yes? And then a human being has matched file names with the sample names, and I found 32 errors in them. I mean, the company was doing analysis on these things. I mean, we live in 21st century. For God's sake don't do anything manual. Hire a programmer and then you normalize [ph?], and then he can write a small software which glues the pieces for you. Stop doing manual.

KP: You mean pay people to do work? <laughs>

Interviewee: What?

KP: I said "Pay people to do work? What is that?"

Interviewee: <laughs> Pharma and biotech, you might think they have so much resources they should have figured it out now. [redacted]. <laughs>

KT: It's getting towards [redacted] o'clock your time, and it's been quite informative and extremely helpful and gives us great insight. A lot of our focus has been on the bench and that side of the development of

gene therapy and genome editing, and this is very good insight into a critical complementary element at the FDA, so we thank you.

Interviewee: It is part of the process. I[redacted].

KT: Right.

Interviewee: Yeah.

KT: [redacted], I assume that was your last-last question?

KP: Yes.

KT: Can't thank you enough for your time. If you have any follow-up questions we'll be happy to answer. If we have follow-up or if we have people who need...

Interviewee: Ask me, and I'll answer them.

KT: As I mentioned, if we have people who need information within your realm of expertise I'd certainly be having them get in touch with you, and then we'll probably be circling back late spring with the redacted transcript. We hope we'll be submitting drafts for publication over the summer, and we'll keep you in the loop as to what we're up to.

Interviewee: Yeah. I have a question for you [ph?]. I have a question. Once you turn this research into a product, I am assuming a publication or some kind of perspective, if you think that might be interesting for FDA regulatory folks I can organize a seminar for you where you can actually give them a perspective and do that. I mean, I think, I mean, you have got freedom of being inspirational about a completely novel technological area and looking at not just science, not just data, not just this, not just that but kind of overall perspective. I think whatever you learn from outside might be helpful for the people from inside, so if at some point in life you realize they are ready for that conversation I can organize a lecture for you, so you'll be invited to FDA to give a lecture.

KT: Oh, well, thank you. That would be great. Are there any people at FDA who you think we might try to speak to? We have some introductions from [redacted], but generally it's challenging to get people who are in FDA to speak on the record while they're at FDA.

Interviewee: Yes. They are barely scripted [ph?], so, I mean, I gave many interviews when I was at the FDA. Usually you have two PR officers sitting next to you. Most of the questions are pre-agreed, and all of the answers are pre-written. It looks so funny. I mean, the two people who would understand only every third word of the scientific word I would pronounce, they were checking if I don't say anything else. <laughs> So it's difficult when you are FDA. It's a whole process. It's kind of long. You're better off getting somebody who left FDA.

KT: That's what we've been doing with fair success.

Interviewee: I can give you contacts, I mean, in [redacted], yes? I mean, one of my good friends, he is now in [redacted]. He used to be the director of [redacted], yes? He was a very nice gentleman, I mean, excellent human qualities, and I think he would be available to kind of give you a perspective into oncology. Obviously I can't connect you. You have to kind of agree with him, yes. I mean, I can give you from that [redacted] a few contacts which are outside now, yes? They might agree [ph?] to give you a perspective. And...

KT: These are the people who you worked with on the [redacted]?

Interviewee: Oh, at FDA in general.

KT: Sure. A couple names would be great.

Interviewee: <overlapping conversation> other [redacted[people in industry. I mean, on the opposite front in a [redacted] kind of...

KT: <laughs> On the [redacted] side we have interest because of some of the other projects we're working on.

Interviewee: Yeah, so...

KT: If any names come across you could shoot me an email.

Interviewee: I'll shoot you a couple of names, and then if you read about them, Google with them and then you are interested I can connect you.

KT: Great. Thank you very much. Have a good weekend.

Interviewee: Thank you.

KT: Hope everyone in the family stays healthy, and we'll follow-up later.

Interviewee: Yes, you too. Just be careful. I see you are in an office. You are not in your house. You are in a risk group.

KT: Me?

Interviewee: Yes.

KT: No, I'm in a high-risk group, but I don't have any comorbidities, which is good, but this is my living room. My wife and I are up in the countryside north of San Francisco and trying to stay away from people as much as we can.

Interviewee: Young people, younger people. <laughs>

KT: <laughs>

Interviewee: Right, [redacted]? <laughs>

KP: Yeah.

KT: Right.

KP: Oh no. I've been walled-off for a month here.

KT: Right.

Interviewee: Yeah. Bye-bye.

KT: Okay.

KP: Bye-bye. Stay safe.

KT: Bye-bye.

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